

This Was Not For Me

Detecting gift purchases as a distinct class of noise in e-commerce recommender systems

AI in Marketing Capstone, Group 2 — [Team Member 1] / [Team Member 2] / [Team Member 3]

github.com/edasaruhan/SIC_AI_17_Capstone_Group_2 · 20 September 2026 · code version 791d9d4

Holding the amount of deleted data fixed, removing gift rows leaves a model that predicts the user's own next purchase better than removing the same number of random rows — and the excess it opens in the recommendation list halves after about one self-purchase.

THE FOUR QUESTIONS

RQ1 · how common are gifts?

19 %

of Toys and Games reviews, after human calibration. 95 % CI [15.8, 22.3]; raw LLM rate 23.2 %. Grocery, the control category, sits at 5.4 %. All four categories peak in December-January (1.58-2.08×).

RQ2 · does removing gifts help?

+15.7% / +40.0%

Recall@10 for C1 – C4 (against the placebo), SASRec / BPR on Toys: +0.005449 [0.0048, 0.0061] and +0.004822 [0.0044, 0.0053]. On the low-gift control it is +1.2% / not detectable — the dose-response holds in all four comparisons.

RQ3 · delete them, or flag them?

delete

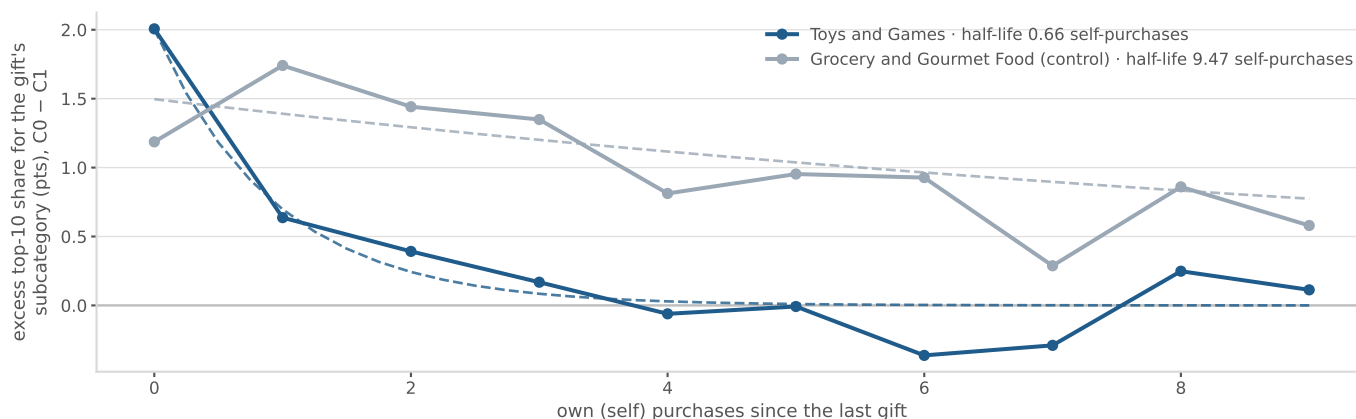
The shadow-token method (C3) was worse than deleting in three of four cells and worse than doing nothing in three of four. Toys · SASRec, C3 – C1 = -0.001235 [-0.0018, -0.0008] → negative. Other flagging designs were not tested.

RQ4 · how long does it last?

0.66

self-purchases until the excess halves, in the sequential model on Toys. 95 % CI [0.53, 0.82]; ≈2.4 weeks (approximate). The gift opens +2.0 pts of the top-10 list, and +4.9 pts of slots go to gift-only subcategories in the classical model.

HOW LONG A GIFT KEEPS PULLING THE LIST · F22, SEQUENTIAL MODEL



WHAT WE CLAIM

- Gift purchases are measurable at scale, and common.
- Gift rows carry less usable preference signal than random rows, with a dose-response across two categories.
- The excess a gift opens decays with the user's own later purchases.
- The one flagging method we tested is worse than deleting.

WHAT WE DO NOT CLAIM

- That the detector is accurate: macro-F1 0.5404 [0.4897, 0.5869] against ONE annotator, and both pre-registered gates (≥ 0.75 , ≥ 0.80) were missed.
- That deleting gifts is a production recommendation — the net effect against doing nothing is model-dependent.
- That this generalises beyond two Amazon categories and two models.
- That wasted slots are lost revenue; conversion was never measured.

HOW IT WAS MEASURED

Qwen3-4B-Instruct-2507 (prompt v3, vllm on 2×T4) labelled 47,200 reviews into a five-class purchase-type schema; one human annotator independently labelled 500 of them; a distilled ModernBERT-base student (fidelity gate PASS, 3/3) then labelled 4,598,612 interactions across the two experiment categories. Six training-set conditions — including two placebos that delete the same NUMBER of random rows — were run through RecBole with SASRec and BPR over 3 seeds, for 72 runs. Gate 1 (may the detector be used?) passed 4/4 in all four categories; Gate 2 (may the contrasts be interpreted?) passed in all four category × model cells. Every contrast is a paired bootstrap over per-user Recall@10, and the interpretation rule — lower bound / not detectable / negative — was written down before any result was seen.